

EDGE-AWARE IMAGE FUSION

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ABSTRACT

This paper presents a comparative analysis of five image fusion methods implemented in a unified computational pipeline. The evaluated techniques include: GraphCut-based segmentation with Poisson and guided filtering refinement, FCN-based segmentation with the same refinement, frequency separation fusion, multi-resolution wavelet fusion, and luminance-transfer fusion. Each method is analyzed in terms of algorithmic approach, strengths, limitations, and appropriate application contexts. The study offers practical insights for choosing suitable strategies across application domains.

Index Terms— Image Fusion, Segmentation, Poisson Blending, GraphCut, Guided Filtering, FCN

1. INTRODUCTION

Image fusion aims to combine complementary information from two images while preserving structural details and avoiding artifacts such as seams or halos. This work analyzes five fusion methods, ranging from traditional frequency-based techniques to advanced segmentation-driven and guided pipelines. We examine their operational principles, failure cases, and ideal use scenarios.

2. METHODOLOGIES AND ANALYSIS

2.1. GraphCut + Poisson Blending + Guided Filtering

Method:

1) GraphCut Segmentation (GrabCut): A bounding box is initialized around the target region, and GrabCut iteratively separates foreground from background by minimizing an energy function $E(L) = U(L) + V(L)$, where U models color likelihoods and V enforces smoothness. This produces a binary mask indicating which pixels should be taken from the sharp image.

2) Poisson Blending: The softened mask is used in Poisson blending, which computes the fused region I by solving $\nabla^2 I = \nabla \cdot (\nabla S)$ subject to boundary constraints from the

colorful image, ensuring smooth gradient transitions and reducing seam artifacts.

3) Guided Filtering: High-frequency details from the sharp image are selectively added back using a guided filter that assumes a local linear model $q_i = a_k p_i + b_k$, where p is the guidance image and q is the filtered output. A gradient-based weight map reinforces edges, yielding crisp and natural transitions.

Limitations: Requires manual initialization. Segmentation struggles if foreground-background contrast is low. It fails when there are multiple objects in the foreground.

Best Use Cases: It works best when the foreground object is high detailed and background has heavy blur or saturation. It's mainly used in interactive editing tasks.

Results: Fig1, Fig2.

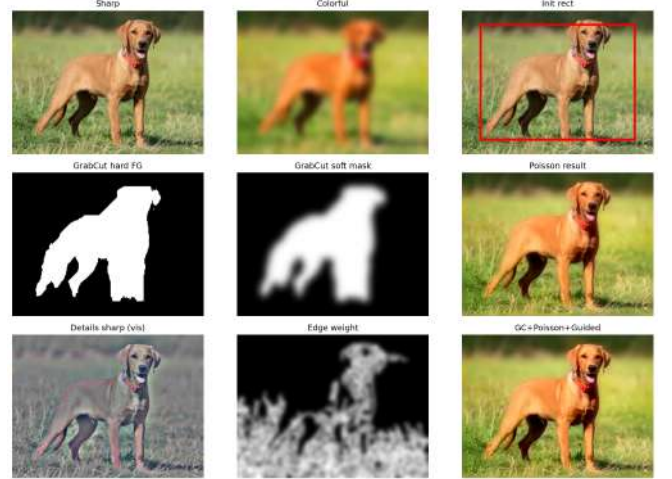


Fig. 1. Stage wise outputs of GraphCut+ Poisson+Guided

2.2. FCN + Poisson + Guided Fusion

Method:

1) FCN-Based Semantic Segmentation: The input image passes through convolution + pooling layers (encoder), which build deep feature maps capturing semantics. This map is then upsampled and converted to labels via argmax over classes at each pixel. This per-pixel label map is used to build

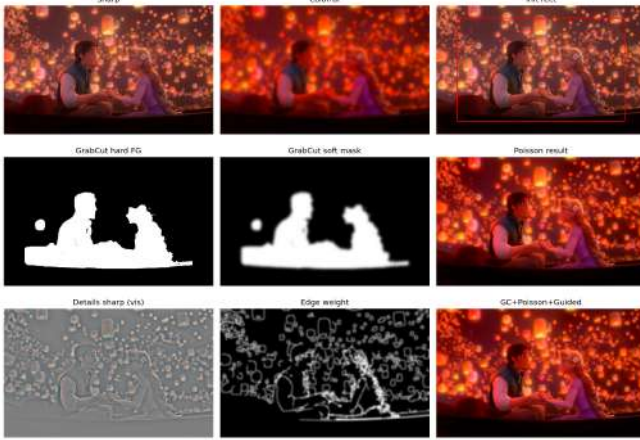


Fig. 2. Stage wise outputs of GraphCut + Poisson + Guided

a foreground mask, which is then softened and used to control where sharp details are injected in the Poisson+guided fusion.

The same **Poisson Blending** and **Guided filtering** from the above method are then implemented.

Limitations: Performance depends on FCN segmentation accuracy and the model's training domain. The pipeline is computationally heavier due to deep network inference.

Best Use Case: It works with multiple objects, complex shapes and overlapping regions due to its automatic, semantics-aware fusion where the network can reliably identify objects. It is ideal for AR workflows and object-guided enhancement. It can also handle uneven exposures.

Results: Fig3, Fig4.

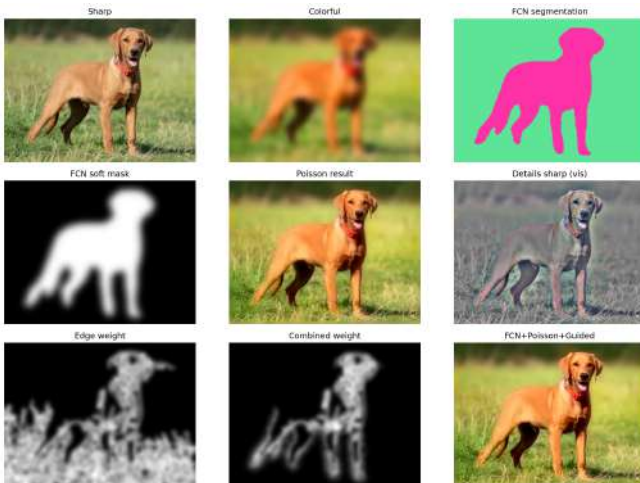


Fig. 3. Stage wise outputs for FCN + Poisson + Guided

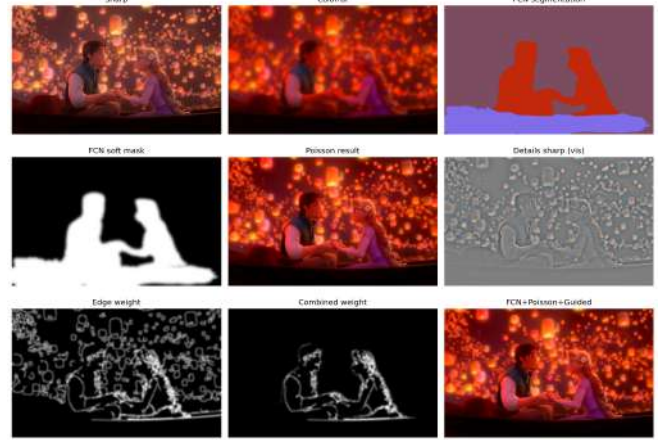


Fig. 4. Stage wise outputs for FCN + Poisson + Guided

2.3. Frequency Separation Fusion

Method:

1) Low-High Frequency Decomposition: Each image is decomposed into a low-frequency layer L (smooth color/structure) and a high-frequency layer H (edges and texture) using Gaussian filtering.

2) Cross-Image Frequency Selection: Fusion is performed by taking the low-frequency component from the colorful image and the high-frequency component from the sharp image.

3) Edge-Preserving Reconstruction: The combined frequency layers are directly summed to reconstruct the final fused RGB image, preserving sharp details while retaining the color characteristics of the colorful image.

Limitations: Assumes a clean separation between detail and color, which does not always hold. May produce color bleeding or loss of mid-frequency structure. Results depend heavily on the Gaussian kernel size.

Best Use Case: Works best for sharp and colorful images. Fast, approximate color-detail merging for stylization, real-time previews, or artistic enhancement where strict accuracy is not required.

Results: Fig5, Fig6.

2.4. Multi-Resolution Wavelet Fusion

Method:

1) Multi-Scale Decomposition: The sharp image is decomposed into a set of detail layers using progressively larger Gaussian kernels, while the colorful image provides the coarse base layer.

2) Multi-Level Detail Transfer: Fusion is achieved by combining the colorful base with the rescaled detail layers from



Fig. 5. Stage wise outputs for Frequency Separation Fusion

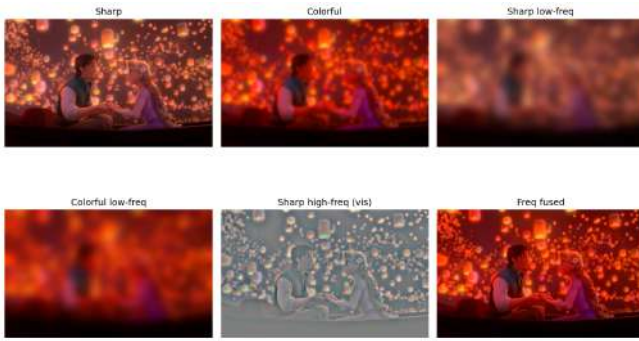


Fig. 6. Stage wise outputs for Frequency Separation Fusion

the sharp image.

3) Final Reconstruction and Enhancement: The fused image is reconstructed by summing all components, and a mild HSV saturation boost is applied to restore color richness after multi-scale filtering.

Limitations: Fixed detail scaling factors may enhance noise or produce haloing. Sensitive to spatial misalignment between inputs.

Best Use Case: Works best for scenes with depth variation or mixed coarse/fine details. Photography-style enhancement and multi-scale detail transfer where controlled sharpening and improved micro-contrast are desired.

Results: Fig7, Fig8, Fig9.

2.5. Luminance Transfer Fusion

Method:

1) Conversion to CIELAB Space: Both images are transformed to the CIELAB color space, separating luminance and chrominance as $I \rightarrow (L, a, b)$, where L captures lightness and (a, b) capture color.

2) Luminance Replacement: The luminance channel of the

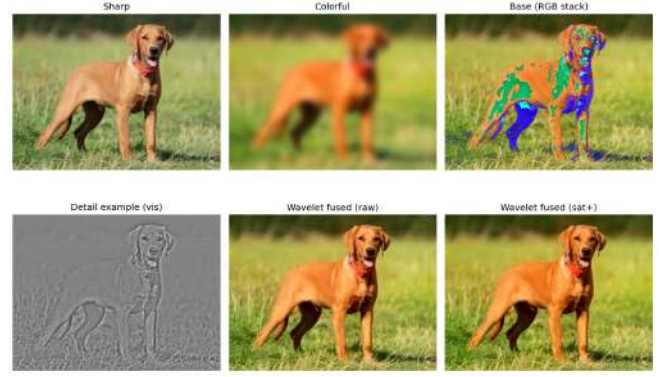


Fig. 7. Stage wise outputs for Multi-Resolution Wavelet Fusion

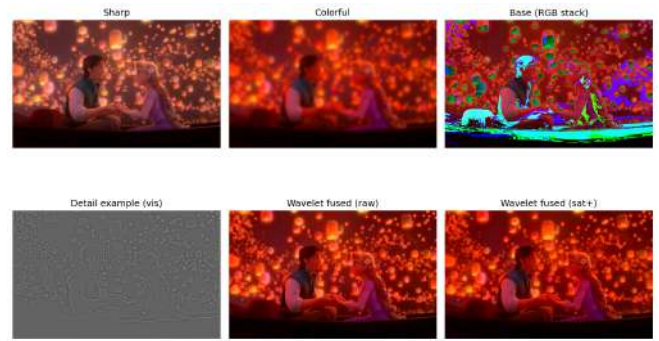


Fig. 8. Stage wise outputs for Multi-Resolution Wavelet Fusion

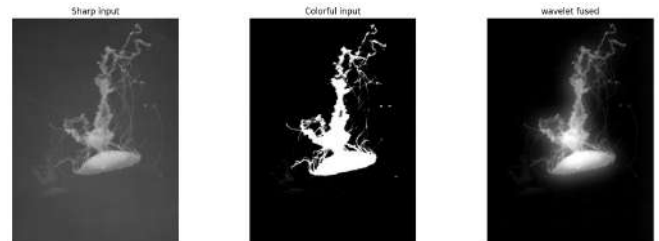


Fig. 9. Example of wavelet producing a halo

colorful image is replaced with that of the sharp image, directly transferring structural brightness information.

3) Reconstruction to RGB: The fused LAB representation (L_f, a_f, b_f) is converted back to RGB, producing an image with the sharp image's structure and the colorful image's chromaticity.

Limitations: Assumes that all structure exists in luminance and all color in chrominance, which often fails. Performs poorly for chromatic edges or correlated color-structure features. Highly sensitive to slight misalignments.

Best Use Case: Works best when one of the inputs has high luminance contrast and the other has high saturation (but low luminance detail). Specialized applications such as grayscale recoloring, illumination replacement, or style-transfer-like tasks where images are perfectly aligned and the luminance–chrominance separation holds. **Results:** Fig10, Fig11, Fig12, Fig13.



Fig. 10. Stage wise outputs for Luminance Transfer Fusion

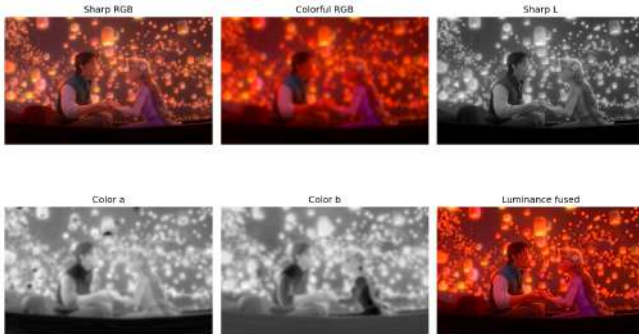


Fig. 11. Stage wise outputs for Luminance Transfer Fusion



Fig. 12. Best case for Luminance Transfer Fusion

3. COMPARATIVE DISCUSSION AND CONCLUSION

The five methods operate across a spectrum of complexity and semantic awareness. Frequency separation and wavelet fusion offer lightweight, global enhancement. Luminance transfer is highly specialized. The GraphCut and FCN-based

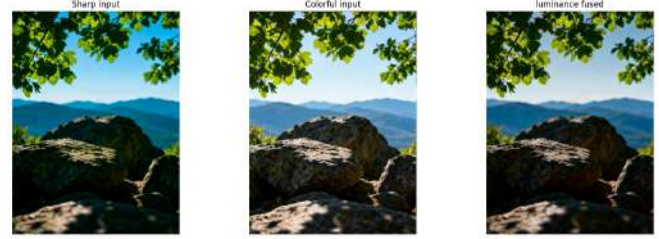


Fig. 13. Stage wise outputs for Luminance Transfer Fusion

pipelines excel at localized, edge-preserving compositing but require greater computational and setup overhead. The choice depends on automation needs, semantic complexity, image alignment, and expected output realism.

3.1. Image 1

- **Sharp input:** Strong structure and fur texture.
- **Colorful input:** Warm, saturated tones but heavily blurred.
- **Freq fused:** Restores detail but produces slight boundary harshness and uneven color mixing.
- **Wavelet fused:** Blends multi-scale texture well but slightly darkens shadows and over-enhances details.
- **Luminance fused:** Maintains natural contrast but introduces mild color shifts in highlights.
- **FCN+Poisson+Guided:** Good structure and color integration but slightly over-smooths uncertain background regions.
- **GC+Poisson+Guided:** Cleanest result with sharp fur, natural saturation, and smooth background blending.

Overall, the best result is GC+Poisson+Guided for its natural look and precise boundaries, while the worst is Freq fused due to uneven color transfer and minor artifacts.

3.2. Image 2

- **Sharp input:** High structural detail but lacking the warm colors of the colorful image.
- **Colorful input:** Rich saturation and glow but heavily blurred with weak structure.
- **Freq fused:** Restores sharpness but introduces halo artifacts and inconsistent color mixing.
- **Wavelet fused:** Improves multi-scale detail but slightly darkens faces and reduces color consistency.



Fig. 14. Image 1

- **Luminance fused:** Strong contrast but suffers from unnatural color shifts due to luminance–chroma mismatch.
- **FCN+Poisson+Guided:** Object-aware and sharp but slightly over-saturated where segmentation is uncertain.
- **GC+Poisson+Guided:** Most natural color–detail balance with clean boundaries thanks to user-guided masking.

Overall, the best result is GraphCut+Poisson+Guided for its clean boundaries and natural fusion, while the worst is the Freq fused output due to haloing and inconsistent color blending.

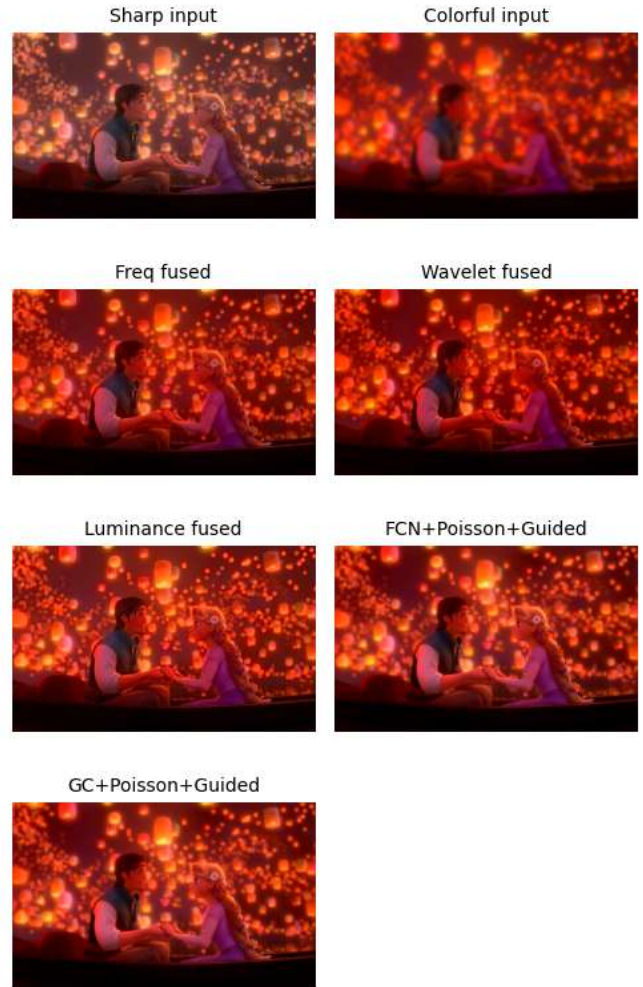


Fig. 15. Image 2

4. REFERENCES

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